

Ansible Inventory Overview

Understanding Static and Dynamic
Inventories with Examples and
Diagrams

What is Ansible Inventory?

- An inventory is a file that defines the list of managed hosts.
- Supports grouping, variables, and plugins.
- Can be static (manual) or dynamic (cloud-based).

Static Inventory Example

More on inventory files

#Sample Inventory File

```
server1.company.com    ansible_connection=ssh  ansible_user=root
server2.company.com    ansible_connection=winrm ansible_user=admin
server3.company.com    ansible_connection=ssh  ansible_ssh_pass=P@#
server4.company.com    ansible_connection=winrm

localhost ansible_connection=localhost
```



Inventory Parameters:

- ansible_connection – ssh/winrm/localhost
- ansible_port – 22/5986
- ansible_user – root/administrator
- ansible_ssh_pass – Password



Security: Ansible Vault

```
server1.company.com
server2.company.com
```

```
[mail]
server3.company.com
server4.company.com
```

```
[db]
server5.company.com
server6.company.com
```

```
[web]
server7.company.com
server8.company.com
```

Dynamic Inventory Example

- Generated using plugins or scripts.
- Useful for cloud environments (AWS, Azure, etc).
- plugin: aws_ec2
- regions:
 - - us-east-1
- filters:
 - tag:Environment: production



What is ansible.cfg?

- **main configuration file** for Ansible.
- It allows you to customize **default behaviors**, such as inventory location, privilege escalation, logging, connection settings, etc.
- Ansible looks for this file in the following order (first found is used):
 1. **ANSIBLE_CONFIG** environment variable (if set)
 2. ./ansible.cfg in the current working directory
 3. ~/.ansible.cfg in the user's home directory
 4. /etc/ansible/ansible.cfg system-wide >>> Redhat

Common Sections and Options

```
[defaults]
inventory = ./hosts
remote_user = ec2-user
host_key_checking = False
retry_files_enabled = False
timeout = 30
```

```
[privilege_escalation]
become = True
become_method = sudo
become_user = root
```

Ansible AD-Hock command

- Ansible-doc command (= default, Author)

Status:

- stableinterface
- preview
- dprecated

Supported_by:

- core
- 3rd party
- community

- Examples for ansible add hock commands
- Diff command and shell module